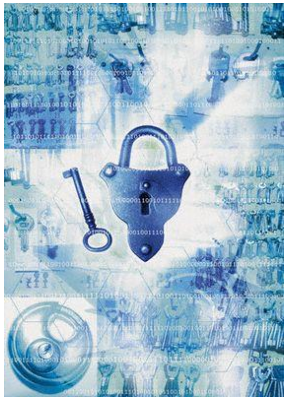
Information Security



Classical cryptography

AI컴퓨터공학부
김 희 열

Characters

- Good guys


Alice


Bob


Charlie

- vs. Bad guys


Trudy


Eve

Security Cornerstones

- Confidentiality
  - To prevent unauthorized reading of information
- Integrity
  - To prevent unauthorized writing of information
- Availability
  - To provide access to information whenever consumers want

Beyond CIA

- Authentication
  - Assurance that the communicating entity is the one claimed
  - Authentication on a stand-alone system
    - Based on cryptography
  - Authentication over networks
    - Based on cryptography and security protocols
- Authorization
  - Enforce restrictions on the actions of authenticated users
    - Based on access control
- Above security mechanisms are implemented in software
  - Bugs
  - Virus, worm
  - OS security

Security Threats

- Threats can come from a range of sources
- Various surveys, with results of order:
  - 55% human error
  - 10% disgruntled employees
  - 10% dishonest employees
  - 10% outsider access
  - also have "acts of god" (fire, flood etc)

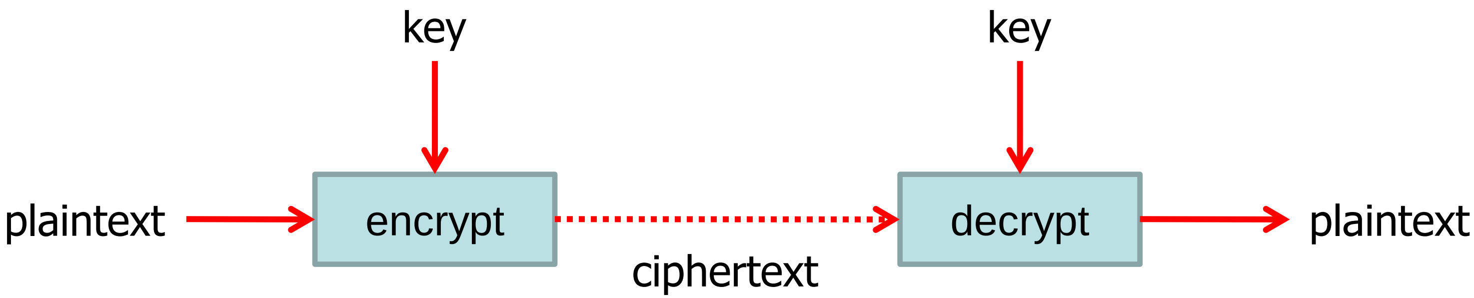
- *Note that in the end, it always comes back to <u>PEOPLE</u>.*
  Technology can only assist so much,
  always need to be concerned about the role of people in the threat equation - who and why.

Crypto

- Cryptology
  - The art and science of making and breaking “secret codes”
- Cryptography
  - Making secret codes
- Cryptanalysis
  - Breaking secret codes
- Crypto
  - All of the above (and more)

How to Speak Crypto

- A cipher or cryptosystem is used to encrypt the plaintext
- The result of encryption is ciphertext
- We decrypt ciphertext to recover plaintext
- A key is used to configure a cryptosystem
- A symmetric key cryptosystem uses the same key to encrypt as to decrypt
- A public key cryptosystem uses a public key to encrypt and a private key to decrypt



Crypto

- Basic assumption
  - The system is completely known to the attacker
  - Only the key is secret
  - Also known as Kerckhoffs Principle
    - Crypto algorithms are not secret

- Why?
  - Secret algorithms are weak when exposed
  - Secret algorithms never remain secret
  - Better to find weaknesses beforehand

Simple Substitution

- Key

Shift by 3!!!

plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

four score and seven years ago

↓

fourscoreandsevenyearsago ←plaintext

↓

IRXUVFRUHDAGVHYHABHDUVDIR ←ciphertext

This is called by “Caesar’s cipher”

Caesar's Cipher Decryption

- Suppose we know the Caesar's cipher is being used
- Ciphertext : IRXUVFRUHDAGVHYHABHDUVDIR

plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- Plaintext : fourscoreandsevenyearsago

Not-so-Simple Substitution

- Shift by n for some $n \in \{ 0,1,\ldots,25 \}$
- The key is n
- Ex) key = 7

plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
ciphertext	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G

Cryptanalysis I : Try Them All

- A simple substitution (shift by n) is used
- But the key is unknown
- Given ciphertext : CSYEVIXIVQMREXIH
- How to find the key?

- Only 26 possible keys → try them all!
- Exhaustive key search

CSYEVIXIVQMREXIH

Key = 1 brxduhwhuplqdwhg

⋮

Key = 4 youareterminated

⋮

- Plaintext : you are terminated

Even-less-Simple Substitution

- The key is some permutation of letters
- Need not be a shift
- Ex)

plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
ciphertext	J	I	C	A	X	S	E	Y	V	D	K	W	B	Q	T	Z	R	H	F	M	P	N	U	L	G	O

- How many keys are possible?
  - Keyspace = 26! > 2^{88}
- If my computer can test 2^{40} keys for a second
  - Exhaustive search needs 4450 millennia on average!

The keyspace should be big enough to make an exhaustive search infeasible

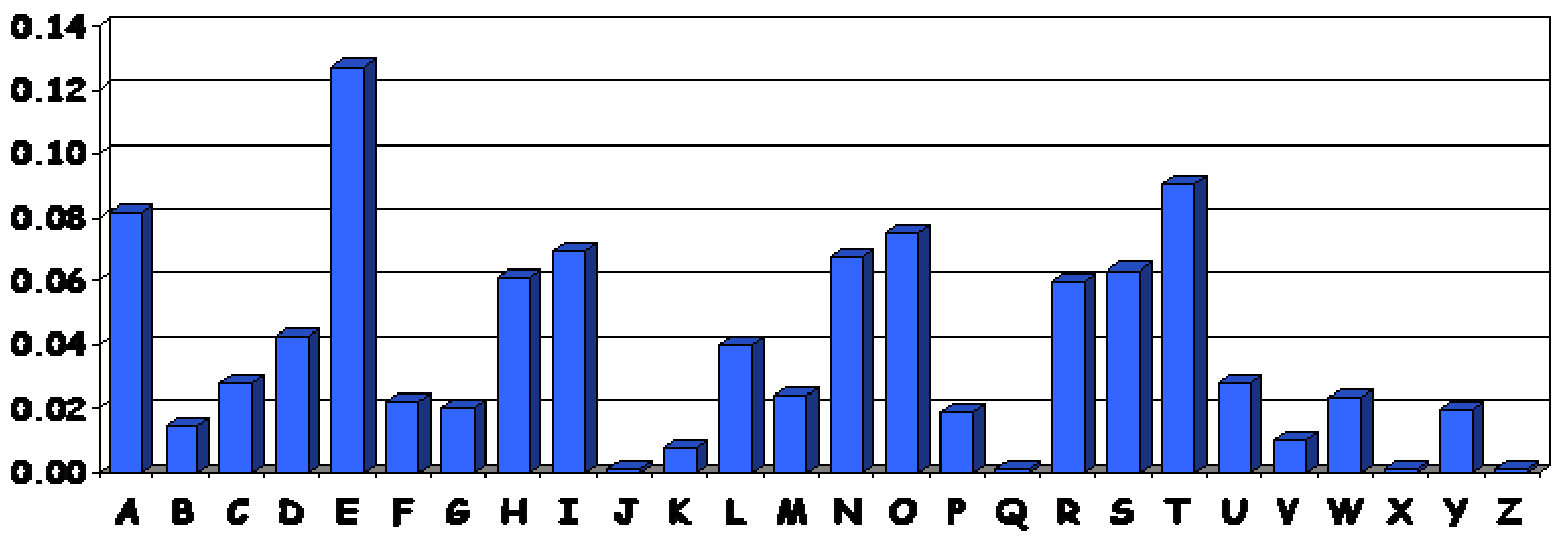
Cryptanalysis II : Be Clever

- We know that a simple substitution is used
- But not necessarily a shift by n
- Can we find the key given ciphertext:

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPQJKT
OYQWIPBVWLXTOXBTFXQWAXBVCXQWAXFQ
JVWLEQNTOZQGGQLFXQWAKVWLXQWAEBIP
BFXFQVXGTVJVWLBTPQWAEBFPBFHCVLXBQ
UFEVWLXGDPEQVPQGVPPBFTIXPFHXZHVFA
GFOTHFEFBQUFTDHZBQPOTHXTYFTODXQH
FTDPTOGHFQPBQWAQJJTODXQHFOQPWTB
DHHIXQVAPBFZQHCFWPFHPBFIPBQWKFABV
YYDZBOTHPBQPQJTQOTOGHFQAPBFEQJHD
XXQVAVXEBQPEFZBVFOJIWFFACFCCFHQWA
UVWFLQHGFXVAFXQHFUFHILTTAVWAFFAWT
EVOITDHFHFQAITIXPFHXAFQHEFZQWGFLVW
PTOFFA

Cryptanalysis II : Be Clever

- Can't try all 2^{88} simple substitution keys
- Can we be more clever?
- Let's use English letter frequency



Cryptanalysis II : Be Clever

- Ciphertext

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPQJKTOYQWIPBVWLXTOXBTFXQWAXBVCXQWAXFQ
JVWLEQNTOZQGGQLFXQWAKVWLXQWAEBIPBFXFQVXGTVJVWLBTPQWAEBFPBFHCVLXBQ
UFEVWLXGDPEQVPQGVPPBFTIXPFHXZHVFAGFOTHFEFBQUFTDHZBQPOTHXTYFTODXQH
FTDPTOGHFQPBQWAQJJTODXQHFOQPWTBDHHIXQVAPBFZQHCFWPFHPBFIPBQWKFABV
YYDZBOTHPBQPQJTQOTOGHFQAPBFEQJHDXXQVAVXEBQPEFZBVFOJIWFFACFCCFHQWA
UVWFLQHGFXVAFXQHFUFHILTTAVWAFFAWTEVOITDHFHFQAITIXPFHXAFQHEFZQWGFLVW
PTOFFA

- Ciphertext frequency counts

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
21	26	6	10	12	51	10	25	10	9	3	10	0	1	15	28	42	0	0	27	4	24	22	28	6	8

e?

Cryptanalysis II : Be Clever

- Ciphertext

PBePVYeBQXZTYePBeEQJHDXXQVAPTPQJKTOYQWIPBVWLXTOXBTeXQWAXBVCXQWAXeQ
JVWLEQNTOZQGGQLeXQWAKVWLXQWAEBIPBeXeQVXGTVJVWLBTPQWAEBePBeHCVLXBQ
UeEVWLXGDPEQVPQGVPPBFTIXPeHXZHVeAGeOTHeEeBQUeTDHZBQPOTHXTYeTODXQH
eTDPTOGHeQPBQWAQJJTODXQHeOQPWTBDHHIXQVAPBeZQHCeWPeHPBFIPBQWKeABV
YYDZBOTHPBQPQJTQOTOGHeQAPBeEQJHDXXQVAVXEBQPEeZBVeOJIWeeACeCCeHQWA
UVWeLQHGeXVAeXQHeUeHILTTAVWAeeAWTEVOITDHeHeQAITIXPeHXAeQHEeZQWGeLVW
PTOeeA

- Ciphertext frequency counts

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
21	26	6	10	12	51	10	25	10	9	3	10	0	1	15	28	42	0	0	27	4	24	22	28	6	8

h? t?

Cryptanalysis II : Be Clever

- Ciphertext

thetVYehQXZTYetheEQJHDXXQVAPTtQJKTOYQWIthVWLXTOXhTeXQWAXhVCXQWAXeQ
JVWLEQNTOZQGGQLeXQWAKVWLXQWAEhItheXeQVXGTVJVWLhTtQWAEhetheHCVLXhQ
UeEVWLXGDtEQVtQGVtthFTIXteHXZHVeAGeOTHeEehQUeTDHZhQtOTHXTYeTODXQH
eTDtTOGHeQthQWAQJJTODXQHeOQtWThDHHIXQVAtheZQHCeWteHthFIthQWKeAhV
YYDZhOTHthQtQJTQOTOGHeQAtheEQJHDXXQVAVXEhQtEeZhVeOJIWeeACeCCeHQWA
UVWeLQHGeXVAeXQHeUeHILTTAVWAeeAWTEVOITDHeHeQAITIXteHXAeQHEeZQWGeLVW
tTOeeA

- Also use the frequency of 2-letter or 3-letter combinations
  - Most frequent 2-letter combinations
    - TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF
  - Most frequent 3-letter combinations
    - THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH
- Try and back steadily until success!

Cryptanalysis II : Be Clever

- Cryptosystem is secure if best known attack requires as much work as an exhaustive key search
  - There is no short-cut attack

- In practice, we must select a cipher that
  - Secure
  - Has a large enough key space

HW #1

- Decrypt the message encrypted using a simple substitution cipher

APS ZU BMS THAAMT KB SOP CHAAPJ MQ LPUWHKX. K UHJ SM JMZ SMLHJ VJ QXKPBLU -- UM PCPB SOMZDO TP QHEP SOP LKQQKEZASKPU MQ SMLHJ HBL SMVMXXMT, K USKAA OHCP H LXPHV. KS KU H LXPHV LPPWAJ XMMSPL KB SOP HVPXKEHB LXPHV. K OHCP H LXPHV SOHS MBP LHJ SOKU BHSKMB TKAA XKUP ZW HBL AKCP MZS SOP SXZP VPHBKBD MQ KSU EXPPL: "TP OMAL SOPUP SXZSOU SM IP UPAQ-PCKLPBS, SOHS HAA VPB HXP EXPHSPL PGZHA." K OHCP H LXPHV SOHS MBP LHJ MB SOP XPL OKAAU MQ DPMXDKH SOP UMBU MQ QMXVPX UAHCPU HBL SOP UMBU MQ QMXVPX UAHCP MTBPXU TKAA IP HIAP SM UKS LMTB SMDPSOPX HS SOP SHIAP MQ IXMSOPXOMML. K OHCP H LXPHV SOHS MBP LHJ PCPB SOP USHSP MQ VKUUKUUKWWK, H USHSP UTPASPXKBD TKSO SOP OPHS MQ KBFZUSKEP, UTPASPXKBD TKSO SOP OPHS MQ MWWXPUUKMB, TKAA IP SXHBUQMXVPL KBSM HB MHUKU MQ QXPPLMV HBL FZUSKEP. K OHCP H LXPHV SOHS VJ QMZX AKSSAP EOKALXPB TKAA MBP LHJ AKCP KB H BHSKMB TOPXP SOPJ TKAA BMS IP FZLDPL IJ SOP EMAMX MQ SOPKX URKB IZS IJ SOP EMBSPBS MQ SOPKX EOHXHESPX. K OHCP H LXPHV SMLHJ. K OHCP H LXPHV SOHS MBP LHJ LMTB KB HAHIHVH, TKSO KSU CKEKMZU XHEKUSU, TKSO KSU DMCPXBMX OHCKBD OKU AKWU LXKWWKBD TKSO SOP TMXLU MQ KBSPXWMUKSKMB HBL BZAAKQKEHSKMB -- MBP LHJ XKDOS SOPXP KB HAHIHVH AKSSAP IAHER IMJU HBL IAHER DKXAU TKAA IP HIAP SM FMKB OHBLU TKSO AKSSAP TOKSP IMJU HBL TOKSP DKXAU HU UKUSPXU HBL IXMSOPXU. K OHCP H LXPHV SMLHJ. K OHCP H LXPHV SOHS MBP LHJ PCPXJ CHAAPJ UOHAA IP PNHASPL, HBL PCPXJ OKAA HBL VMZBSHKB UOHAA IP VHLP AMT, SOP XMZDO WAHEPU TKAA IP VHLP WAHKB, HBL SOP EXMMRPL WAHEPU TKAA IP VHLP USXHKDOS, HBL SOP DAMXJ MQ SOP AMXL UOHAA IP XPCPHAPL HBL HAA QAPUO UOHAA UPP KS SMDPSOPX.

HW #1

- Hint : substitution cipher

- Upload a pdf file via LMS system including
  - Used program code
  - Frequency count table, Key table
  - Plaintext decrypted

- Due : before next class

Double Transposition Cipher

- Plaintext : attack at dawn

	col 1	col 2	col 3
row 1	a	t	t
row 2	a	c	k
row 3	x	a	t
row 4	x	d	a
row 5	w	n	x

Permute rows and columns

	col 1	col 3	col 2
row 3	x	t	a
row 5	w	x	n
row 1	a	t	t
row 4	x	a	d
row 2	a	k	c

- Ciphertext : xtawxnattxadakc
- Key : matrix size and permutations (3,5,1,4,2) and (1,3,2)
- Do not substitute letters

Use the combination of substitution and transposition!

One-time Pad

- A provably secure cryptosystem

letter	e	h	i	k	l	r	s	t
binary	000	001	010	011	100	101	110	111

Encryption : Plaintext ⊕ Key = Ciphertext

	h	e	i	l	h	i	t	l	e	r
Plaintext:	001	000	010	100	001	010	111	100	000	101
Key:	111	101	110	101	111	100	000	101	110	000
Ciphertext:	110	101	100	001	110	110	111	001	110	101
	s	r	l	h	s	s	t	h	s	r

One-time Pad

- XOR(⊕) operation
  - $0 \oplus 0 = 0$
  - $0 \oplus 1 = 1$
  - $1 \oplus 0 = 1$
  - $1 \oplus 1 = 0$

- Property
  - Identity element : $A \oplus 0 = A$
  - Self-inverse : $A \oplus A = 0$

One-time Pad

- Decryption

letter	e	h	i	k	l	r	s	t
binary	000	001	010	011	100	101	110	111

Decryption : Ciphertext ⊕ Key
= Plaintext ⊕ Key ⊕ Key = Plaintext

	s	r	l	h	s	s	t	h	s	r
Ciphertext:	110	101	100	001	110	110	111	001	110	101
Key:	111	101	110	101	111	100	000	101	110	000
Plaintext:	001	000	010	100	001	010	111	100	000	101
	h	e	i	l	h	i	t	l	e	r

One-time Pad

- Double spy claims the sender used “fake key” :
  101 111 000 101 111 100 000 101 110 000

letter	e	h	i	k	l	r	s	t
binary	000	001	010	011	100	101	110	111

	s	r	l	h	s	s	t	h	s	r
Ciphertext:	110	101	100	001	110	110	111	001	110	101
“fake key”:	101	111	000	101	111	100	000	101	110	000
“Plaintext”:	011	010	100	100	001	010	111	100	000	101
	k	i	l	l	h	i	t	l	e	r

One-time Pad

- Sender is captured and claims the key is:
  111 101 000 011 101 110 001 011 101 101

	s	r	l	h	s	s	t	h	s	r
Ciphertext:	110	101	100	001	110	110	111	001	110	101
"Key":	111	101	000	011	101	110	001	011	101	101
"Plaintext":	001	000	100	010	011	000	110	010	011	000
	h	e	l	i	k	e	s	i	k	e

One-time Pad

- Provably secure when used correctly
  - Ciphertext provides no information about plaintext
  - Pad must be random, used only once (why?)

    $C1 = P1 \oplus K$, $C2 = P2 \oplus K$
    Trudy knows (P1, C1), then
    $(C1 \oplus C2) \oplus P1 = (P1 \oplus K \oplus P2 \oplus K) \oplus P1 = P1 \oplus P2 \oplus P1 = P2$

- But, it's impractical
  - How to distribute the secret key?

Real-world One-time Pad

- Project VENONA
  - Soviet spy messages from U.S in 1940's about Nuclear bombs
  - Spy carried one-time pad into U.S
  - Spy used pad to encrypt secret messages
  - Repeats within the one-time pads made cryptanalysis possible

> [C% Ruth] learned that her husband [v] was called up by the army but he was not sent to the front. He is a mechanical engineer and is now working at the ENORMOUS [ENORMOZ] [vi] plant in SANTA FE, New Mexico. [45 groups unrecoverable]
>
> detain VOLOK [vii] who is working in a plant on ENORMOUS. He is a FELLOWCOUNTRYMAN [ZEMLYaK] [viii]. Yesterday he learned that they had dismissed him from his work. His active work in progressive organizations in the past was cause of his dismissal. In the FELLOWCOUNTRYMAN line LIBERAL is in touch with CHESTER [ix]. They meet once a month for the payment of dues. CHESTER is interested in whether we are satisfied with the collaboration and whether there are not any misunderstandings. He does not inquire about specific items of work [KONKRETNAYa RABOTA]. In as much as CHESTER knows about the role of LIBERAL's group we beg consent to ask C. through LIBERAL about leads from among people who are working on ENOURMOUS and in other technical fields.

Codebook

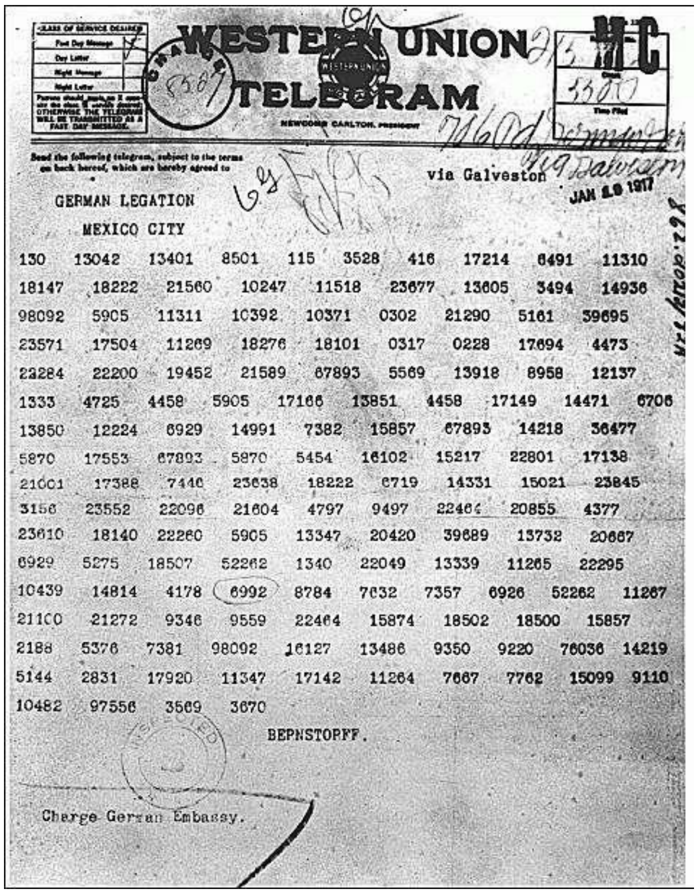
- A book filled with "codewords"

Februar	13605
fest	13732
finanzielle	13850
folgender	13918
Frieden	17142
Friedenschluss	17149
:	:

- Word-based complex substitution cipher

Zimmerman Telegram

- One of most famous codebook ciphers
- Led to US entry in WWI
- British had recovered partial codebook



WESTERN UNION TELEGRAM

NEWCOMB CARLTON, PRESIDENT

Send the following telegram, subject to the terms on back hereof, which are hereby agreed to

via Galveston

JAN 19 1917

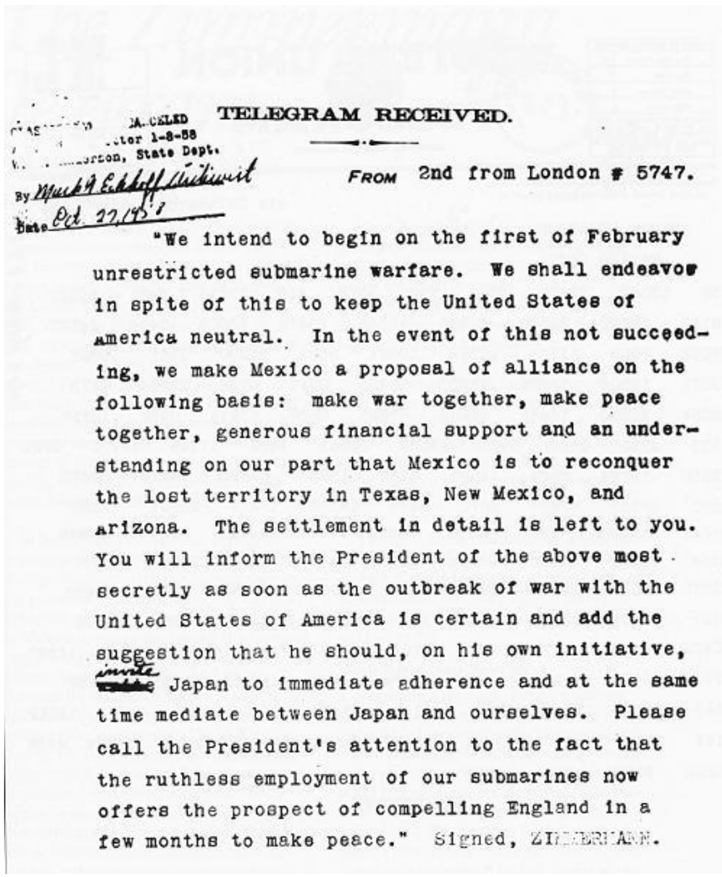
GERMAN LEGATION

MEXICO CITY

130 13042 13401 8501 115 3528 416 17214 6491 11310
18147 18222 21560 10247 11518 23677 13605 3494 14936
98092 5905 11311 10392 10371 0302 21290 5161 39695
23571 17504 11269 18276 18101 0317 0228 17694 4473
22284 22200 19452 21589 67893 5569 13918 8958 12137
1333 4725 4458 5905 17166 13851 4458 17149 14471 6706
13850 12224 6929 14991 7382 15857 67893 14218 36477
5870 17553 67893 5870 5454 16102 15217 22801 17138
21001 17388 7446 23638 18222 6719 14331 15021 23845
3156 23552 22096 21604 4797 9497 22464 20855 4377
23610 18140 22260 5905 13347 20420 39689 13732 20667
6929 5275 18507 52262 1340 22049 13339 11265 22295
10439 14814 4178 6992 8784 7632 7357 6926 52262 11267
21100 21272 9346 9559 22464 15874 18502 18500 15857
2188 5376 7381 98092 16127 13486 9350 9220 76036 14219
5144 2831 17920 11347 17142 11264 7667 7762 15099 9110
10482 97556 3569 3670

BERNSTORFF.

Charge German Embassy.



TELEGRAM RECEIVED.

FROM 2nd from London # 5747.

"We intend to begin on the first of February unrestricted submarine warfare. We shall endeavor in spite of this to keep the United States of America neutral. In the event of this not succeeding, we make Mexico a proposal of alliance on the following basis: make war together, make peace together, generous financial support and an understanding on our part that Mexico is to reconquer the lost territory in Texas, New Mexico, and Arizona. The settlement in detail is left to you. You will inform the President of the above most secretly as soon as the outbreak of war with the United States of America is certain and add the suggestion that he should, on his own initiative, invite Japan to immediate adherence and at the same time mediate between Japan and ourselves. Please call the President's attention to the fact that the ruthless employment of our submarines now offers the prospect of compelling England in a few months to make peace." Signed, ZIMMERMANN.

Recent History of Crypto

- Claude Shannon (1940’s)
  - Father of the science of information theory
- Computer revolution
  - Need for handling lots of data
- Data Encryption Standard (DES) (1970’s)
- Public key cryptography (1970’s)
- Advanced Encryption Standard (AES) (1990’s)
- Modern cryptography (2000’s)

- Crypto moved out of classified world

Claude Shannon

- The founder of Information Theory
- 1949 paper : Information Theory of Secrecy Systems

- Confusion and diffusion
  - Confusion
    - Obscure relationship between plaintext and ciphertext
    - Ex) Substitution ciphers, one-time pad
  - Diffusion
    - Spread plaintext statistics through the ciphertext
    - Ex) Double transposition cipher
  - Proved that one-time pad is secure

Taxonomy of Cryptography

- Symmetric key
  - Same key for encryption as for decryption
  - Stream ciphers
  - Block ciphers

- Public key
  - Two keys: one for encryption (public key), one for decryption (private key)
  - Digital signature

- Hash algorithms

Taxonomy of Cryptanalysis

- Ciphertext only attack
  - Basic and always possible
- Known plaintext attack
  - Trudy knows some previous (plaintext, ciphertext) pairs
  - Use stereotypical headers s.t. email header
- Chosen plaintext attack
  - Lunchtime attack
- Adaptively chosen plaintext attack
  - “adaptive” is important
- Etc…

favorable
to attacker